

# PRINCIPLE OF MANAGEMENT OF CLASS-I MALOCCLUSION

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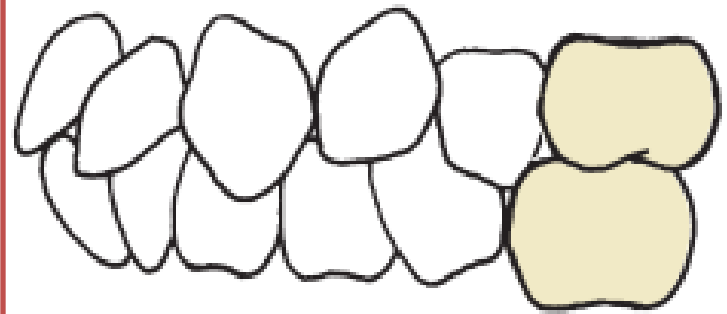
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# LEARNING OBJECTIVES

- *EXPLAIN DIFFERENT PRINCIPLES OF MANAGEMENT OF CLASS-I, CLASS-II AND CLASS-III MALOCCLUSIONS*

# Introduction

- Class I malocclusion, as defined by Angle, incorporates a normal inter-arch relationship.
- Class I malocclusion is said to exist when there is a harmonious relationship of the underlying skeletal structures and the malocclusion component is restricted to the dental malrelations only.
- These malocclusions may typically include individual tooth malalignments and/or malocclusions in the vertical or transverse planes.
- The most commonly encountered malocclusions in this group are the bimaxillary protrusion/proclination and cases involving crowding in the maxillary and mandibular arches



Class I malocclusion

## ➤ Mal-occlusions encountered in class-I malocclusions include

- *Bidentoalveolar protrusion*
- *Crowding*
- *Spacing*
- *Midline diastema*
- *Rotations*
- *Missing tooth/teeth*
- *Supernumerary teeth*
- *Extrusion/intrusion of tooth/teeth*
- *Transposition*
- *Linguoversion/Buccoversion*
- *Impaction*

# Associated problems include:

- ✓ Lack of oral hygiene maintenance
- ✓ Periodontal problems
- ✓ Unpleasant Facial esthetics-primary concern of the patient
- ✓ Problems associated with teeth or the surrounding soft tissues

# FEATURES OF CLASS I SKELETAL MALOCCLUSION:

## **Skeletal features**

- Harmonious face
- Straight to convex profile
- Nothing really abnormal

# Dental features

- Molar relation is Angles Class I
- Individual tooth malocclusions may be seen with varying degree of severity(peg laterals, missing teeth, supernumerary teeth/mesodens, rotation, crowding, midline diastema, spacing, proclined incisors...)
- Malocclusion may be present in the vertical and transverse planes(crossbites, openbites, deepbites)
- Lip competence is dependent on the degree of anterior proclination

# A.BIMAXILLARY DENTOALVEOLAR PROTRUSION

- The most common malocclusion in Angle's class I
- Due to proclined upper and lower anterior teeth



# Dental features

- Dental bimaxillary proclination
- Decreased interincisal angles
- Spacing between teeth may be seen
- Normal molar and canine relationship

**What else**

# Management:

Fixed orthodontic appliance therapy with premolars extraction approach and followed by retraction of anterior teeth.

**How about at early ages?  
Lip exercise and habit  
breakers**

## B. MIDLINE DIASTEMA

- Refers to an anterior midline spacing between the two maxillary/mandibular central incisors
  - Any spacing or gaps existing in the midline of the dental arch
  - Most commonly seen in maxillary arch
  - **???Most frequently seen malocclusion**
  - Considered as easy to treat but often difficult to retain
- Normal: in deciduous dentition
- Abnormal: in permanent dentition



# **Etiology:**

- **Transient malocclusion**
  - Generalized spacing during deciduous dentition
  - Ugly-duckling stage during mixed dentition
  - Abnormal frenal attachment
  - ALTM discrepancy: missing teeth, true/relative microdontia, macrognathia, peg shaped laterals

- Abnormal pressure habits: tongue thrusting, thumb sucking, mouth breathers
- Trauma: tooth loss, fracture of teeth
- Heredity-as familial trait
- Racial prediction: Negroid race
- Midline pathology: cyst, tumors, odontoms, mesodens
- Iatrogenic: RPE as prognostic sign
- Periodontal migration of teeth

- True Macroglossia
- Flaccid lip muscles
- Imperfect fusion at the midline of pre-maxilla: Moyers
- Improper dental alignment
- Anterior traumatic bite: TFO
- Rotations of CIs around 90 degree from normal
- Bolton's discrepancy

# Diagnosis:

- Proper history and clinical examination
- Blanch test
- Radiographs (notching appearance, missing teeth, mesodens, pathologies)
- Identification of abnormal habits
- Model analysis: to determine TMAL discrepancy
- Presence of gingival recession in lower anterior

# Management:

➤ The treatment of midline diastema is done in three phases:

Phase I: removal of the cause

Phase II: active treatment

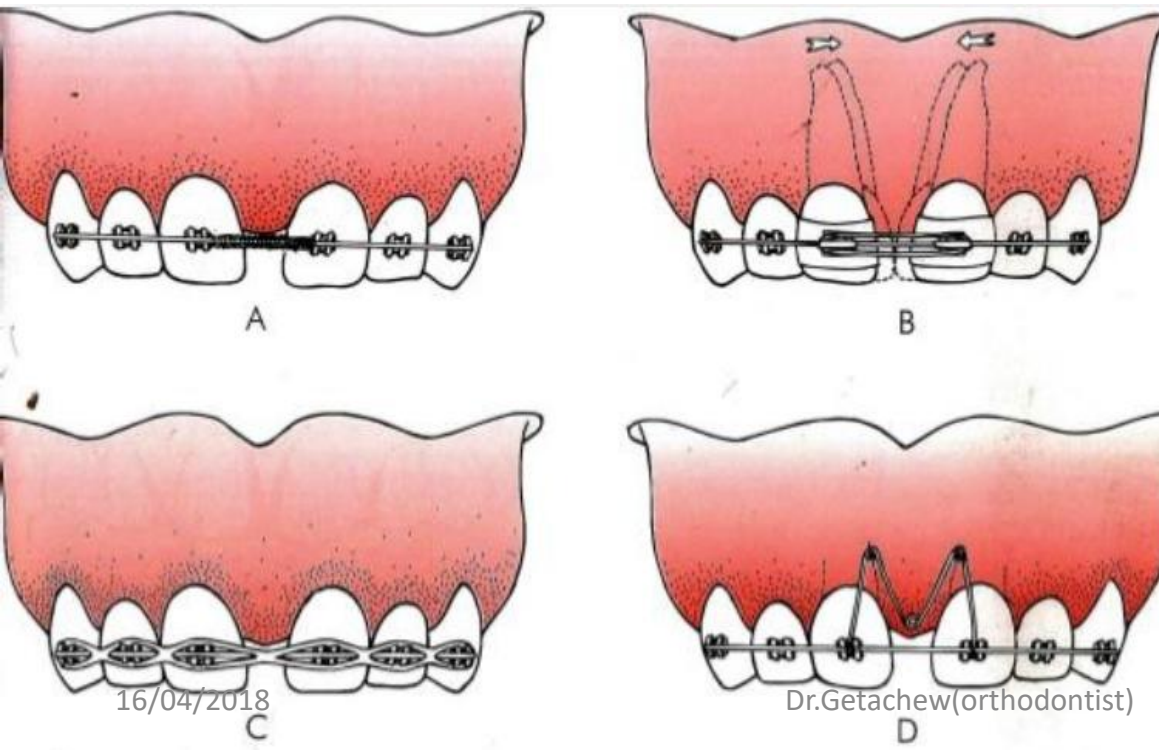
Phase III: retentive phase

# 1. Phase I: Removal of The Cause:

- a. Removal of habits: removable or fixed habit breakers: tongue rake/crib, oral screen
- b. Extraction of unerupted mesodens
- c. Frenectomy with/without gingivoplasty???**
- d. Treatment of any midline pathology
- e. Extraction of supernumerary teeth before midline diastema closure

## 2. Phase II: Active Treatment:

- a. Removable orthodontic appliances therapy with finger springs, split labial bows, active short labial bows
- b. Fixed orthodontic appliances therapy with closed coil spring, elastic thread, e-chain, "M" shaped springs
- c. Implants: replacement of missing teeth
- d. **Cosmetic restorations**/prosthesis (peg laterals: Jacket crowns, missing teeth: implants)



# **3. Phase III: Retentive Phase:**

Long term retention using suitable retainers

**1. Lingual or palatal bondable retainers**

2. Hawley's retainers

3. Post-treatment composite build-ups

Before tx



After tx



## C. SPACING

- The presence of spacing between the teeth is one of the commonly seen manifestation of a class I malocclusion
- Normal in deciduous dentition: a positive prognostic sign
- Abnormal in permanent dentition
- Classified as: localized or generalized



# Etiology:

1. Disproportion between arch length and tooth material:
  - *Oligodontia, Microdontia, Macroglossia*
2. Peg shaped laterals
3. Deleterious oral habits: *Thumb sucking, Tongue thrusting, Mouth breathing*
4. Macroglossia
5. Presence of impacted supernumerary teeth
6. Pathological or cystic lesions between teeth
7. Premature loss of permanent teeth

# Diagnosis:

- History and clinical examination
- Model analysis: ALTM discrepancy
- Radiographic examination: Bony pathology, Impacted teeth

# Treatment:

## I. Removal of the etiology

- Removal of abnormal pressure habits using habit breakers
- Removal of bony pathological/cystic lesions
- Extraction of supernumerary teeth
- Removal of un-erupted tooth by open extraction

## **II. Use of removable or fixed appliances**

- Active ROA therapy using active labial bows
- FOA therapy using e-chains, elastic threads

### **III. Use of crowns and prosthesis**

- Using suitable Jacket crowns for microdontia tooth, Peg shaped laterals
- Composite build-ups

### **IV. Implants, FPD or RPD:**

- For those missing teeth

## D. CROWDING

- It is another common manifestation of a class I malocclusion
- Occurs as a result of disproportion between tooth size and arch length: ALTM discrepancy



# Etiology:

- ALTM discrepancies: micrognathia, macrodontia
- Presence of supernumerary teeth
- Prolonged retention of deciduous teeth with impaction or ectopically erupted successors teeth
- Abnormalities in size and shape of teeth: localized or generalized macrodontia
- Premature loss of deciduous teeth with drifting of adjacent teeth in to the extraction space

# Diagnosis:

- History and clinical examination
- Model analysis
- Identification and determination of the cause/etiology
- Radiology: retained tooth, impaction, congenitally missing teeth

# Management:

- Gaining space
- Use of removable appliances: ROA incorporate coil springs, canine retractors, labial bows
- Use of FOA with multi-looped arch wires, superelastic NiTi arch wires

## E. ROTATIONS

- “Rotations are tooth movements that occur around their long axis”
- Two types: ML/P or DB/L and MB or DL/P
- They have a very high risk of relapse due to the stretching of the supraalveolar and transeptal gingival fibers which readapt very slowly to the new position



# **Etiology:**

- Any discrepancy/barrier in path of eruption
- Prolonged retained deciduous tooth
- Pathology, cyst
- Abnormal orthodontic forces

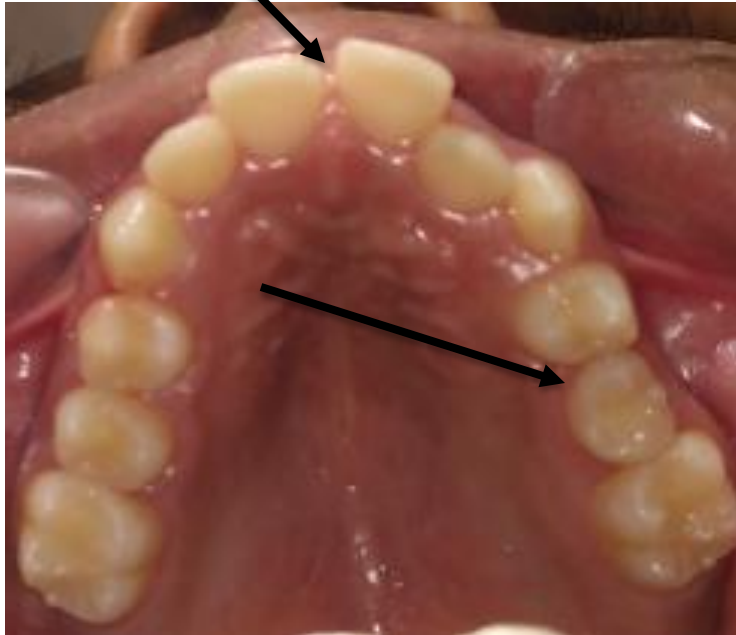
# Diagnosis:

- Proper history and clinical examination
- Model analysis
- Radiology

# Management:

- Space management
- Use of ROA incorporating “Z” spring, double cantilever spring
- FOA: rotation wedges, derotation spring, couple force, resilient NiTi arch wires
- Retention: CSCF (Circumferential supra-crestal fibrotomy), retainers

BEFORE

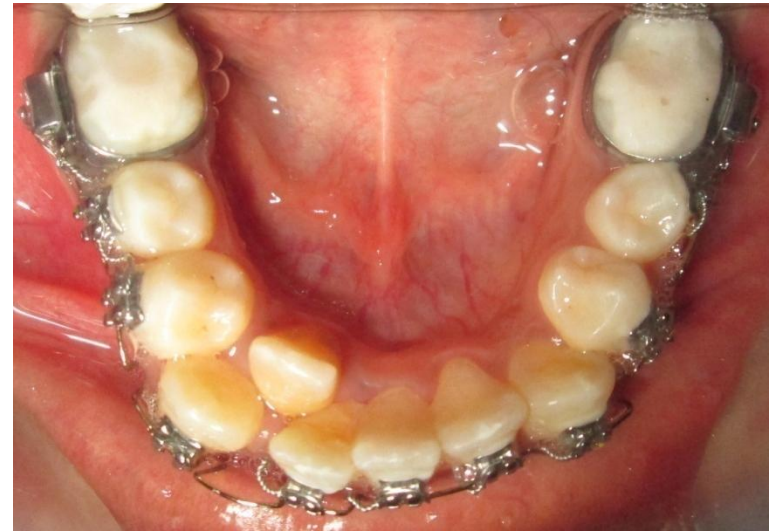


AFTER





29/02/2014 1<sup>st</sup> day of wire engagement  
14ss wire with multiloops on both archs

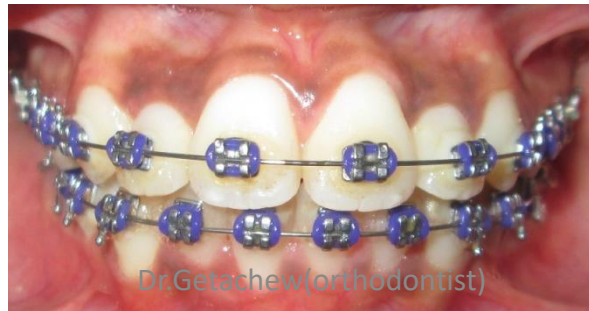


16/04/2018

Dr.Getachew(orthodontist)

38

10/10/2014 4<sup>th</sup> month appointment  
16NiTi on lower and 18 NiTi on upper arch archs



# PRINCIPLE OF MANAGEMENT OF CLASS-II MALOCCLUSION

# Introduction

- Depending on classification of malocclusions by Angle:
  - Angle's class II malocclusion is characterized by a class II molar relationship where:
  - Mandibular arch is in a distal relation to that of maxillary arch
- “The DBC of the maxillary first permanent molar occludes in the buccal groove of the mandibular first permanent molar”



- Further classified as:
  1. Angle's class II division 1 malocclusion:
    - Class II molar relationship with proclined maxillary anterior teeth
  2. Angle's class II division 2 malocclusion:
    - Class II molar relationship with retroclined maxillary anterior teeth



# Angle class-II div-I malocclusion

- Clinical features:

1. Class II molar relationship (either end-on or full-fledged class II)
2. Proclined maxillary anteriors
3. Increased overjet
4. Convex facial profile
5. Excessive curve of Spee
6. Short and hypotonic upper lip
7. Lip trap
8. Lack of anterior lip-seal
9. Proclined, extruded mandibular anteriors
10. Abnormal muscle activity:



Convex

11. NLA: Too much acute

12. Skeletal features:

- Maxillary protrusion/prognathism
- Mandibular retrusion/retrognathism
- Maxillary prognathism and Mandibular retrognathism

# Etiology

- Broadly, the etiological factors can be classified as:
  1. Prenatal
  2. Natal and
  3. Postnatal

## 1. Prenatal Factors:

- Heredity: prognathic maxilla/ retrognathic mandible/both
- Teratogenesis: teratogenic agents(drugs, x-rays exposure, infections....)
- Intra-uterine fetal posture
- Accidents

## 2. Natal Factors:

- Trauma to condylar region due to improper application of forceps during pregnancy
- Fibrosis or ankylosis of TMJ
- Underdevelopment of mandible
- Mandibular deficiency/retrognathism

### 3. Postnatal Factors:

- Traumatic injury to the mandible and TMJ
- Long-term irradiation therapy of the skeletal craniofacial region
- Infectious conditions: Rheumatoid arthritis can influence mandibular growth
- Abnormal habits:
  - Mouth breathing
  - Thumb sucking
- Low tongue position lead to unbalanced muscle activity: hyperactive buccinators group of muscles

# Treatment objectives

- Ideal overjet and overbite: reduction of overjet and overbite
- Correction of crowding, irregularities of teeth
- To establish stable incisor, canine and molar relationships: correction of unstable molar relationship
- Correction of posterior crossbite
- Normalizing the musculature

# Diagnosis

- History, clinical examination
- Model analysis
- Photographs: intraoral/extraoral
- Radiographs
- Lateral cephalograms

# Classification

## A. Skeletal:

1. Prognathic maxilla with Orthognathic mandible
2. Orthognathic maxilla with Retrognathic mandible
3. Prognathic maxilla with Retrognathic mandible

## B. Dental:

1. Class II Div. 1
2. Class II Div. 2
3. Class II Div. 1/2, subdivision on the right/left side

There are three basic approaches to the treatment of class II division 1 malocclusion(skeletal):

1. Growth modification
2. Camouflage
3. Surgical correction

## A. Growth Modification **early mixed dentition :**

- Functional
- Orthopaedic appliances

### ➤ **I. To correct maxillary prognathism:**

- head gear to restrict further maxillary growth



## ➤ II. To correct mandibular deficiency/retrognathism:

a. During mixed dentition period:

- Removable myofunctional appliance therapy, such as
  - Activator
  - Functional regulator



b. During the end of growth period:

- Fixed functional appliances like

- Herbst appliance

- Jasper jumper



C. To correct both maxillary prognathism and mandibular retrognathism/deficiency:

- Activator with headgear to restrict maxillary growth and promote mandibular growth

## B. Camouflage:

- Camouflage/masking by orthodontic tooth movement by:
  - ❑ Extraction of certain teeth and moving the rest of the teeth in to space created
  - ❑ Distalization of maxillary molars prior to eruption of second molars
  - ❑ The underlying skeletal discrepancy is not addressed

## C. Surgical Correction:

- In patients with severe skeletal malrelationship
- After the completion of growth
  1. Maxillary setback: to correct maxillary prognathism
  2. Mandibular advancement: to correct mandibular retrognathism

# Angle class-II div-II malocclusion

## ➤ Clinical features:

1. Class II molar relationship
2. Retroclined maxillary anteriors
3. Pleasing straight facial profile
4. Deepbite (increased overbite)
5. Broad square face
6. Backward path of closure
7. Deep mento-labial sulcus
8. Absence of abnormal muscle activity
9. Risk of trauma of periodontium of upper palatal and lower labial aspects
10. Square lower arch form

# Treatment objectives

1. Relief of gingival trauma
2. Correction of incisal relationship
3. Relief of crowding and local irregularities
4. Correction of class II molar relationship

# Treatment protocol

1. Correction of incisal relationship/ Correction of deep bite:

□ The deepbite is reduced by use of:

- Anterior bite plane
- Fixed orthodontic appliances incorporating anchor bends (intrusion arches), reverse curve of Spee

2. Alteration of incisal inclination:

Use of torqueing springs to move the upper incisors (roots palatally, crowns labially)

# PRINCIPLE OF MANAGEMENT OF CLASS-III MALOCCLUSION

# Definition

- ❑ *“The MBC of the maxillary first permanent molar occludes in the interdental space between the mandibular first and second permanent molars”*
- ❑ Mandible is in a mesial relation to the maxillary arch, mesiocclusion
- ❑ A malocclusion that is very easy to identify but is often difficult to treat
- ❑ Highest incidence in Japan and Korea



# Classification

1. Dental
  - .Class III subdivision
2. Skeletal/ True Class III
3. Pseudo class III

# Clinical Features Of Class III Malocclusion

- ✓ Class III molar relationship
- ✓ Incisors relationship:
  - Normal
  - Edge-to-edge or
  - Anterior crossbite (reverse overjet)
- ✓ Narrow or short upper arch, broad and long lower arch
- ✓ Posterior crossbite (a common feature of class III)
- ✓ Crowding in upper arch, spacing in lower arch(dental)
- ✓ Prominent chin
- ✓ Concave facial profile (due to presence of prominent chin or prognathic mandible)

- ✓ In vertical growers:
  - Increased intermaxillary height: anterior openbite
  - In some patients: deepbite
  - Reverse overbite
- ✓ Anterior crossbite with reverse overbite
- ✓ **In pseudo class III patients:** characterized by
  - Presence of occlusal prematurities/discrepancies
  - Resulting habitual forward positioning of mandible while closing; these patients exhibit
  - Forward path of closure

# **Skeletal Features Of Class III Malocclusion:**

- ✓ A short or retrognathic maxilla
- ✓ A long or prognathic mandible
- ✓ A combination of the above

# Etiology

## 1. True class III malocclusion:

- Inherited skeletal imbalance
- Very strong genetic basis

## 2. Pseudo class III malocclusion:

- Occlusal prematurities
- Enlarged adenoids .....result  
habitual forward positioning of  
mandible

# Diagnosis:

- ❖ History, clinical examination
  - *Molar, canine, incisors relationships*
  - *Facial profile, divergence, clinical FMA*
  - *Observation of path of closure*
  - *Anterior crossbite*
- ❖ Model analysis
- ❖ Photographs: intraoral/extraoral
- ❖ Radiographs
- ❖ Lateral cephalograms

# Flowchart For Treatment Of Class III Malocclusion

## A.IN GROWING PATIENT

### **1. Skeletal Class III**

- ✓ For maxillary retrognathism: face mask to protract maxilla
- ✓ For mandibular prognathism with maxillary retrognathism: face mask with chin cup or myofunctional appliance
- ✓ For mandibular prognathism: chin cup to restrict mandibular overgrowth

### **2. Dental class III: fixed orthodontic treatment**

## **B.NON-GROWING PATIENT**

**1. Dental class III:** fixed orthodontic treatment

**2. Skeletal class III:**

**a. Severe class III**

✓ Due to mandibular prognathism: Surgical mandibular set back

✓ Due to maxillary retrognathism: Surgical maxillary advancement

**b. Mild to moderate class III:**

✓ Orthodontic camouflage by extraction of some teeth: only in lower arch

✓ Fixed orthodontic treatment

# Management

## ***I. INTERCEPTION DURING GROWTH***

- Growth modulation procedures include:

1. Frankel III myofunctional appliance

2. Chin cup with high pull head gear: for mandibular

prognathism to inhibit mandibular overgrowth

3. Reverse head gear (Face mask): for maxillary retrognathism to protract maxilla

## II. TREATMENT OF SEVERE CLASS-III AFTER GROWTH COMPLETION

- Surgical or corrective procedures include:
  - 1. Maxillary advancement procedures:**
    - ✓ To treat class III malocclusion due to maxillary deficiency/retrognathism
    - ✓ Le Fort I osteotomy
  - 2. Mandibular set back procedures:**
    - ✓ To treat class III malocclusion due to mandibular prognathism

### ***III. TREATMENT OF PSEUDO CLASS-III***

- ✓ Removal of occlusal discrepancies/prematurities
- ✓ Removal of the cause/etiology

### ***IV. TREATMENT OF ANTERIOR CROSS-BITE***

- ✓ Lower anterior inclined planes: Catalan's appliance
- ✓ Removable appliance incorporating screws for anterior expansion

### ***V. TREATMENT OF POSTERIOR CROSS BITE***

- ✓ Can be treated by RPE,
- ✓ Segmental expansion

# Role Of Extractions In Class III

## **A. For lower arch length deficiency and anterior cross bite:**

- FOA therapy with lower first premolar extraction approach
- Class III intermaxillary elastics; to retract lower anteriors

## **B. For arch length deficiency in both arches:**

- FOA therapy with lower and upper first premolar extraction approach
- Avoid upper arch extraction approach